Akash Prajapati

Python Backend Engineer | Applied AI & LLM Systems

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PROFESSIONAL SUMMARY

Python Backend Engineer with 3.4+ years of experience developing scalable backend systems, REST APIs, and microservices using Python, Django, DRF, and FastAPI. Experienced in database optimization, asynchronous processing, and enterprise integrations, with hands-on exposure to Applied AI, RAG pipelines, LLM integrations, and semantic search across automotive, e-commerce, and AI-driven applications.

TECHNICAL SKILLS

Programming & Backend: Python, SQL, Django, Django REST Framework, FastAPI, REST APIs, Microservices

Applied AI & GenAI: LLMs, RAG, Prompt Engineering, LangChain, LangGraph, Tool Calling, OpenAI APIs, Semantic Retrieval, Image Embeddings, Sentence Transformers, ChromaDB, Whisper, ElevenLabs, Guardrails, LLM Harnessing, LLM Evaluation, LangSmith

Databases & Messaging: PostgreSQL, Redis, Django ORM, Celery, Apache Kafka

Authentication & Integrations: Keycloak, JWT, OAuth 2.0, OTP Verification, Payment Gateway Integration, Third-Party API Integration, RBAC

Concepts: OOP, Database Design, Query Optimization, Agile Methodology, System Design

Cloud & Tools: AWS (S3, RDS, EC2, ELB), Docker, Kubernetes (Fundamentals), Grafana, Git, GitHub, Swagger/OpenAPI

PROFESSIONAL EXPERIENCE

Vishleshan Software Solutions Pvt. Ltd. | Software Developer 25th November 2025 – 24th September 2026

Edukaan — Tata Motors | Commercial Vehicle Spare Parts E-Commerce

- Developed 100+ REST APIs using Python, Django, and DRF for core e-commerce workflows on a platform handling 1,000+ orders daily.
- Implemented Smart Search using separate image and text embedding pipelines, enabling product discovery through images, part numbers, names, categories, and voice queries, with top-5 similarity-based results.
- Developed FastAPI notification services integrated with Apache Kafka, supporting approximately 1,000–10,000 notifications daily.
- Implemented Redis caching for Order History APIs and optimized Django ORM queries to reduce repeated database access and improve response times.
- Built asynchronous report and export workflows using Celery and Redis, integrating AWS S3 for file storage and secure presigned downloads.
- Developed DSR/MSR (Field Representative) business workflows and integrated Shiprocket for shipment creation and tracking, and SAP Grow with internal backend services.

Bhasasetu — Whirlpool | Hindi Voice-Enabled AI Assistant

- Developed backend APIs and an end-to-end RAG pipeline using Python and Django to deliver context-grounded responses from a product knowledge base.
- Implemented document processing, recursive chunking (500 words per chunk, 20-word overlap), Sentence Transformers embeddings, and ChromaDB vector storage for approximately 100–500 knowledge-base items.
- Implemented semantic retrieval using cosine similarity and a dynamic threshold, orchestrating retrieval and response generation through LangGraph workflows with conditional routing, query rephrasing, and fallback handling; integrated Whisper, GPT-4, and ElevenLabs for speech-to-text, response generation, and text-to-speech.
- Implemented LLM safety controls, including PII redaction, structured output validation, and prompt injection detection; integrated LangSmith for execution tracing, debugging, and monitoring workflow latency and token usage. Evaluated response quality across ~5,000 test cases using expected answers and LLM-based evaluation, achieving an average score of 87%.

PV Shopee — Tata Motors | Passenger Vehicle Spare Parts E-Commerce

- Integrated Keycloak authentication APIs for login and token-generation workflows using JWT and OAuth 2.0.
- Implemented CAPTCHA validation and supported backend maintenance and production issue resolution.

Sankey Solutions | Solution Analyst 1st June 2023 – 14th November 2025

FleetEdge — Tata Motors | IoT-Based Fleet Management Platform

- Developed backend features across vehicle, wrapper, subscription, and payment services within an approximately 18-microservice fleet management platform using Python, Django, and PostgreSQL.
- Built the Vehicle Day-wise Tracker backend end-to-end, processing historical GPS coordinates, calculating distance using the Haversine formula, identifying trips and stops, and preparing route data for Google Maps visualization.

- Designed and optimized backend services, improving API performance by 20-30% by identifying and resolving N+1 query issues and optimizing database access using Django ORM Subquery and related query optimizations.
- Implemented role- and subscription-based access controls, managing Fleet Owner and Fleet Manager permissions, site-level vehicle visibility, and plan-based access to vehicle data and fleet management modules.
- Developed Mileage Saarathi APIs and business logic to process driving parameters such as speed, idling, harsh acceleration, braking, and gear shifting/RPM to generate mileage-improvement guidance.
- Implemented historical telemetry synchronization through Sahaj APIs, using 30-minute cron scheduling, checkpoint-based incremental fetching, backlog recovery, and PostgreSQL upserts keyed by VIN and timestamp.
- Integrated CCAvenue payment gateway with request encryption, response decryption, payment callbacks, status updates, subscription synchronization, and Redis cache updates; also developed vehicle subscription workflows for plan purchases, upgrades, renewals, and expiry.

IQMS — Intelligent Quality Management System | Tata Motors Passenger Vehicle Manufacturing

- Developed the User Portal backend using Python, Django, and PostgreSQL, covering database design and implementation of 50+ REST APIs to digitize paper-based vehicle assembly-line quality inspections.
- Implemented Quality Maker and Quality Checker workflows for inspection recording, defect reporting, corrective actions, and verification across assembly-line operations.
- Developed an Apache Kafka consumer to receive VIN and vehicle information from manufacturing sites and persist incoming data in PostgreSQL; integrated Swagger/OpenAPI documentation for API testing and frontend coordination.

Vidura — AI-Assisted Git Repository Change & Security Analysis

- Independently developed an AI-assisted code analysis solution integrating GitHub APIs to retrieve commit differences and surrounding code context.
- Implemented separate OpenAI API workflows for code-change summarization and potential security vulnerability detection.
- Developed checks for exposed API keys, access tokens, passwords, email credentials, and AWS access details, generating structured reports with issue descriptions and severity levels.

EDUCATION

Bachelor of Engineering — Computer Engineering
Smt. Indira Gandhi Engineering College, University of Mumbai | Aug 2019 – Jun 2023 | CGPA: 8.88/10

ACHIEVEMENTS

- **Shining Star:** Recognized for outstanding technical contributions.
- **Bar Raiser:** Recognized for problem-solving, technical guidance, and supporting team members.

CERTIFICATIONS

- Machine Learning with Google Cloud
- Android Development

PUBLICATION

- Crop Protection Against Birds Using Deep Learning and IoT, IRJET, Vol. 10, Issue 4, April 2023.